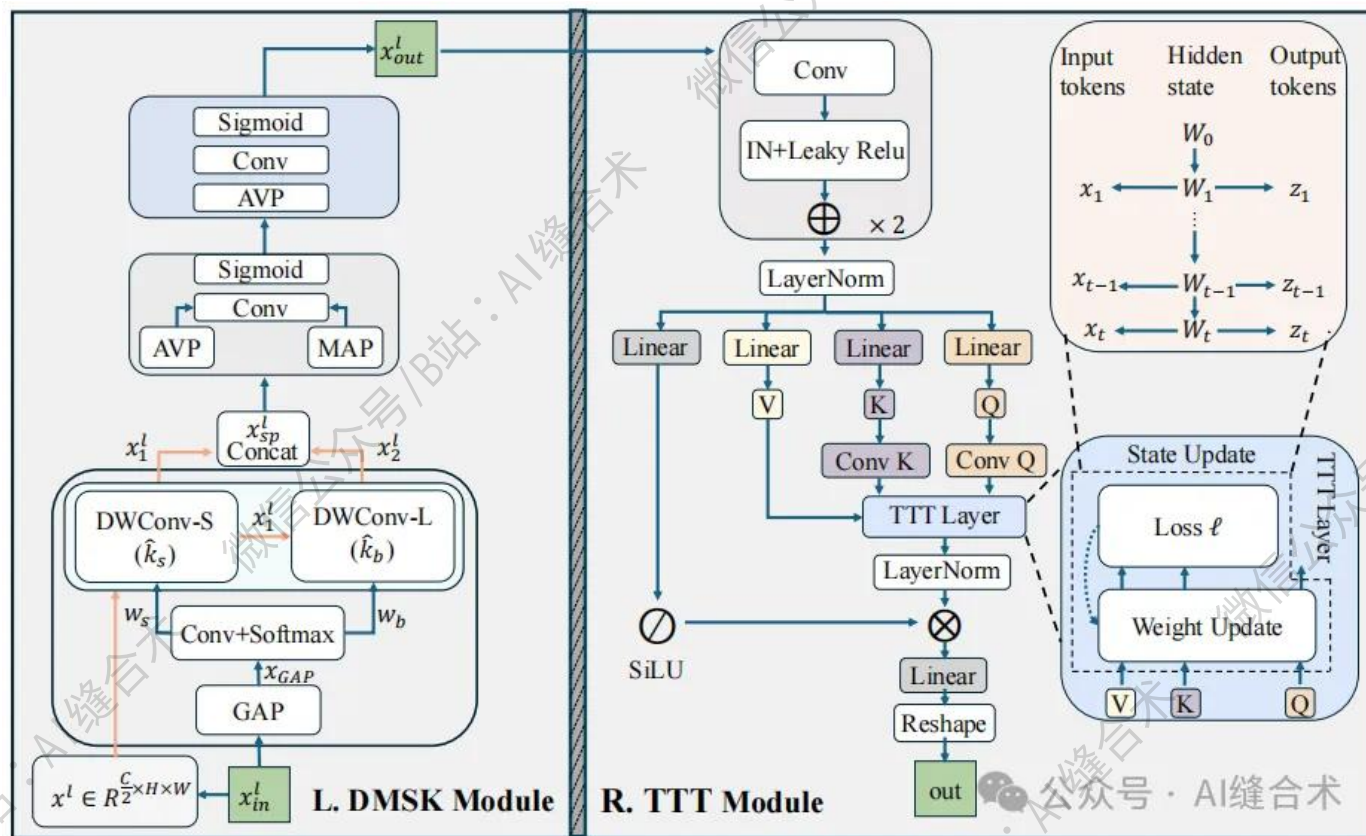


核心速览



论文题目: Enhancing Feature Fusion of U-like Networks with Dynamic SkipConnections

中文题目: 通过动态跳跃连接增强类 U 网络的特征融合

论文链接:

<https://arxiv.org/pdf/2509.14610>

所属单位: 四川大学

核心速览:

为解决 U 型网络中传统跳跃连接存在的特征间约束（静态融合）和特征内约束（多尺度交互不足）问题，提出动态跳跃连接（DSC），通过测试时训练（TTT）模块和动态多尺度核（DMSK）模块增强跨层连接的自适应性，并在多种 U 型网络架构上验证其有效性。

<https://mp.weixin.qq.com/s/7Cil0QGRURs63Kr8tFwIYA>

以下代码仅提供 DMSK 模块：

```
DMSKModule(  
    (proj_1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1))  
    (act): GELU(approximate='none')  
    (spatial_gating_unit): DMSK(  
        (channel_proj): Conv2d(64, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)  
        (dynamic_kernel_selection): DynamicKernelSelection(  
            (conv_layers_1): ModuleList(  
                (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32)  
                (1): Conv2d(32, 32, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=32)  
            )  
            (conv_layers_2): ModuleList(  
                (0): Conv2d(32, 32, kernel_size=(7, 7), stride=(1, 1), padding=(9, 9), dilation=(3, 3), groups=32)  
                (1): Conv2d(32, 32, kernel_size=(9, 9), stride=(1, 1), padding=(12, 12), dilation=(3, 3), groups=32)  
                (2): Conv2d(32, 32, kernel_size=(11, 11), stride=(1, 1), padding=(15, 15), dilation=(3, 3), groups=32)  
            )  
            (attention_1): Sequential(  
                (0): AdaptiveAvgPool2d(output_size=1)  
                (1): Conv2d(32, 2, kernel_size=(1, 1), stride=(1, 1))  
                (2): Softmax(dim=1)  
            )  
            (attention_2): Sequential(  
                (0): AdaptiveAvgPool2d(output_size=1)  
                (1): Conv2d(32, 3, kernel_size=(1, 1), stride=(1, 1))  
                (2): Softmax(dim=1)  
            )  
        )  
        (spatial_se): Sequential(  
            (0): Conv2d(2, 2, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3))  
            (1): Sigmoid()  
        )  
    )  
    (proj_2): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1))  
)  
input_tensor_shape : torch.Size([2, 64, 32, 32])  
output_tensor_shape : torch.Size([2, 64, 32, 32])
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！